

The Anatomy of a Modern Breakup: A Multi-Dataset Sentiment and Engagement Analytics Pipeline

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Abstract.

This paper presents a multi-dataset sentiment and engagement analytics pipeline applied to Reddit posts from r/breakups and r/relationship_advice. Using VADER sentiment scoring, Latent Dirichlet Allocation (LDA) topic modeling, and three supervised classification models – logistic regression, decision tree, and random forest – we investigate what structural and textual features drive community engagement in online relationship support communities. Our findings confirm the Sentiment Paradox : despite the highly emotional nature of breakup narratives, textual sentiment shows near zero correlation with post engagement. Instead, comment volume and posting hour are the dominant engagement predictors. To address class imbalance in the binary engagement target, we applied class weighting and Synthetic Minority Oversampling Technique (SMOTE), improving recall for high-engagement posts from a baseline of 24% to 94%. Five recurring relational themes were identified through LDA: Heartbreak and Recovery, Long-term Strain, Dating and Early Relationships, Intimacy and Dynamics, and Social and Family Conflict. These results demonstrate that structural interaction mechanics, rather than emotional content, are the primary determinants of online peer-support visibility.

Keywords : sentiment analysis, engagement prediction, Reddit, natural language processing, LDA topic modeling, class imbalance, class weighting, SMOTE, random forest, logistic regression, online communities

I. INTRODUCTION

Online relationship support communities on Reddit – most notably “r/breakups” and “r/relationship_advice” – host thousands of unique submissions daily, generating large-scale unstructured text datasets that capture the granular evolution of interpersonal distress. These platforms have become primary avenues for individuals navigating romantic dissolution, offering peer support that supplements or replaces traditional counselling. [3]

Despite the scale of these communities, no systematic computational workflow exists to categorize relationship narratives into thematic clusters or to identify the structural drivers behind community engagement. Traditional NLP approaches focus predominantly on emotional sentiment, operating under the implicit assumption that emotionally charged content naturally attracts more community attention.

This paper challenges that assumption. We present an integrated analytics pipeline that combines VADER sentiment scoring, LDA topic modeling, and supervised binary classification to determine what truly drives post visibility in online relationship communities. Our central contribution is the empirical confirmation of the Sentiment Paradox - the finding that textual emotional valence is essentially irrelevant to community engagement [2], while structural features such as comment volume and posting timing are dominant predictors of thread visibility [5].

The remainder of this paper is organized as follows : Section II reviews related work, Section III describes our methodology, Section IV presents experimental results, Section V discusses findings and implications, and Section VI concludes with directions for future work.

II. RELATED WORK

The study of online peer-support communities has gained significant momentum within computational social science. Prior work has established that Reddit serves as a primary avenue for processing relational distress, with communities exhibiting distinct linguistic and behavioral patterns compared to general social media platforms [3].

Sentiment analysis of social media text has transitioned to specialized tools. Noviaristanti [7] evaluated VADER’s utility on social platforms, demonstrating its optimization for informal text, while recent studies reinforce its strength in extracting sentiment from unstructured public discourse [2].

The relationship between emotional content and engagement is debated. Hubner et al. [5] demonstrated that Reddit engagement is governed by structural posting characteristics and readability ease rather than simple content triggers. Furthermore, applying LDA to Reddit support forums has successfully identified recurring themes in distress and coping mechanisms [3].

Topic modeling of relationship discourse has been explored in mental health contexts. Prior implementations using LDA on Reddit data have successfully identified recurring themes in depression and anxiety communities, providing methodological precedent for our application to relationship distress narratives.

Class imbalance is a persistent challenge. Gupta et al. [4] reviewed modern applications of SMOTE and class-weighting, showing that handling data skew is vital for text classification. Similarly, modern pipelines frequently leverage oversampling to stabilize predictors on imbalanced human-behavior datasets [1, 10].

III. METHODOLOGY

A. Dataset

We combined two publicly available Kaggle datasets : `relationship_advice.csv` (1,885 rows, 8 columns) from `r/relationship_advice` [9] and `reddit_breakup_dataset_cleaned.csv` (948 rows, 10 columns) from `r/breakups` [8]. After merging with schema standardization and a source flag column, the combined corpus totaled 2,833 posts. Following data cleaning, 2,825 posts were retained for analysis.

B. Data Preprocessing

Schema alignment was the first preprocessing step, resolving column name inconsistencies across source files (e.g., `upvotes` vs. `score`, `post_date` vs. `timestamp`). Timestamps were parsed using mixed format handling to accommodate both `YYYY-MM-DD` and `DD/MM/YYYY` formats present across the two datasets.

Text normalization included lowercasing, URL removal via regex, punctuation stripping, and whitespace collapsing applied to both title and body fields. Eight posts (0.3%) with empty body text post-cleaning were removed. For topic modeling, an additional 825 posts containing automated moderation content were filtered using regex-based keyword matching, yielding a clean LDA [1] corpus of 2,000 posts.

C. Feature Engineering

Five features were engineered for classification modeling: `body_length` (word count of cleaned body text), `publish_hour` (hour extracted from timestamp, 0-23), `title_sentiment` (VADER compound score for post title), `body_sentiment` (VADER compound score for post body), and `comms_num` (total comment volume). The binary target variable `high_engagement` was created via median split (median score = 1.0), yielding a 68.2% / 31.8% class distribution.

D. Sentiment Analysis

VADER sentiment scoring was applied to both title and body text, generating compound polarity scores ranging from -1 (most negative) to +1 (most positive). Each post was categorized as positive (score ≥ 0.05), negative (score ≤ -0.05), or neutral. Posts with body sentiment above the 0.05 threshold were classified as positive.

E. Topic Modeling

Latent Dirichlet Allocation (LDA) was applied to the filtered 2,000-post corpus using scikit-learn's `LatentDirichletAllocation` with `n_components=5` and `max_iter=20`. Text was vectorized using `CountVectorizer` with `max_df=0.90`, `min_df=10`, `max_features=1000`, and an extended stopword list combining NLTK English stopwords with 60 domain-specific Reddit noise terms.

F. Classification Models

Three binary classifiers were evaluated : Logistic Regression (with `StandardScaler` preprocessing), Decision Tree (`max_depth=5`), and Random Forest (`n_estimators=100`, `max_depth=5`). All models were trained on an 80/20 stratified train/test split (2,260 training / 565 test rows). Three training strategies were applied : baseline (default parameters), class-weighted (`class_weight='balanced'`), and SMOTE [4] (synthetic minority oversampling to achieve 50/50 class balance, yielding 3,084 training rows).

IV. EXPERIMENTS AND RESULTS

A. Sentiment Analysis Results

VADER scoring revealed that 57% of posts (1,613) scored positive sentiment, 38% (1,087) scored negative, and 5% (125) were neutral. The mean body sentiment was +0.17, indicating a surprising positive bias in breakup content. Correlation analysis showed `body_sentiment` correlation with `high_engagement` of -0.02 and `title_sentiment` correlation of -0.04 — confirming near-zero predictive relationship between emotional tone and engagement.

B. Topic Clustering Results

LDA identified five coherent thematic clusters:

Topic	Label	Proportion
1	Heartbreak & Recovery	30.4%
2	Long-term Strain	21.3%
3	Dating & Early Relationships	17.9%
4	Intimacy & Dynamics	16.2%
5	Social & Family Conflict	14.4%

The dominance of Heartbreak & Recovery (30.4%) reflects the `r/breakups` community composition, while Long-term Strain (21.3%) captures the `r/relationship_advice` demographic of established relationship issues.

C. Baseline Classification Results

Table II presents baseline model performance on the 20% holdout test set:

Model	Accuracy	F1	Recall
Logistic Regression	0.72	0.34	0.23
Decision Tree	0.70	0.29	0.19
Random Forest	0.71	0.35	0.24

While accuracy reached 70-72%, recall for high-engagement posts remained critically low (19-24%), indicating that baseline models defaulted to predicting the majority class due to class imbalance [10]. This confirmed the need for class imbalance correction.

D. Improved Model Results

Table III shows results after applying `class_weight='balanced'`:

Model	Accuracy	F1	Precision	Recall
LR (Balanced)	0.699	0.560	0.524	0.600
DT (Balanced)	0.646	0.630	0.472	0.944
RF (Balanced)	0.658	0.615	0.480	0.856

Table IV shows SMOTE results:

Model	Accuracy	F1	Precision	Recall
LR (SMOTE)	0.710	0.582	0.538	0.633
DT (SMOTE)	0.683	0.634	0.502	0.861
RF (SMOTE)	0.660	0.610	0.481	0.833

Class weighting (penalizing majority class bias) [10] produced the most dramatic recall improvements, with Decision Tree (Balanced) achieving 0.944 recall – a 4.98x improvement over baseline. SMOTE offered a more balanced precision-recall tradeoff [4], with Decision Tree (SMOTE) achieving 0.861 recall with 0.502 precision.

E. Feature Importance

Random Forest feature importance analysis ranked `comms_num` as the dominant predictor (importance = 0.50), followed by `publish_hour` (0.20) and `body_length` (0.13). Critically, `body_sentiment` (0.09) and `title_sentiment` (0.08) ranked lowest. Logistic Regression coefficient analysis corroborated this ordering, with `comms_num` coefficient magnitude of 1.40 versus `title_sentiment` of 0.03.

V. DISCUSSION

A. The Sentiment Paradox

The near-zero correlation between emotional sentiment and community engagement (-0.02 for `body_sentiment`) challenges the dominant assumption in social media analytics that emotionally charged content naturally attracts more attention. In the context of peer-support communities, our findings suggest users engage with posts for functional rather than purely emotional reasons – they respond to posts where they can offer concrete advice, share similar experiences, or validate specific situations, regardless of how emotionally positive or negative the post is [3].

This finding aligns with the Jobs-to-be-Done framework, which conceptualizes community participation as fulfilling specific functional or social needs rather than passive emotional resonance. The

unexpected positive sentiment bias (+0.17 mean) further suggests that even breakup narratives are predominantly framed with resolution-seeking rather than pure distress expression.

B. Structural Determinism in Engagement

The dominance of `comms_num` (importance 0.50) as an engagement predictor is both expected and important. Comment volume is partially circular – posts that get commented on get more visibility, which leads to more comments. This feedback loop reflects Reddit's ranking algorithm behavior, where early interaction velocity drives subsequent visibility [5]. The `publish_hour` feature (importance 0.20) captures the temporal dimension of this dynamic: posts published during peak community activity windows receive early interaction that seeds the visibility loop.

These findings have direct practical implications: optimizing posting timing and seeding early interaction (through strategic community participation) are more effective levers for visibility than crafting emotionally compelling content.

C. Class Imbalance and Recall

The improvement in recall from 24% to 94% through class weighting demonstrates that baseline model underperformance was entirely attributable to class imbalance [10] rather than feature inadequacy. The 68/32 class distribution, while appearing manageable, was sufficient to cause systematic majority-class bias in all three baseline models. Both class weighting and SMOTE successfully corrected this, with class weighting producing superior recall at the cost of precision, and SMOTE offering a more balanced tradeoff.

VI. CONCLUSION AND FUTURE WORK

This paper presented a complete analytics pipeline for online relationship support community data, combining sentiment analysis, topic clustering, and iterative classification modeling. The central finding – the Sentiment Paradox – establishes that emotional tone has negligible predictive power for community engagement in peer-support forums. Instead, structural features (comment volume and posting timing) are the dominant determinants of post visibility.

The modeling pipeline demonstrated measurable improvement through class imbalance correction, raising recall from 24% to 94% using class weighting. Five coherent relational themes were identified through LDA, providing a structured taxonomy of online relationship distress.

Future work should explore semantic embedding models (BERT, RoBERTa) to capture nuanced contextual meaning beyond surface-level polarity scoring. Multi-class engagement labeling (Low / Medium / Viral) would provide finer-grained prediction targets. Time-series modeling of engagement velocity – capturing how quickly a post gains its first interactions – could provide earlier and more actionable triage signals for community moderators.

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